Year 7

REVISION PACK

Unit 3

Name: _____

Class: _____

Unit 3 · Collecting like terms

Worked example: simplifying expressions

Simplify, by collecting like terms.

(a) $6x + 2x - x$

(b) $-3p + 7p - 2p$

(c) $5x + 4y - 2x - y$

(d) $a^2 + 4a + 2a^2 - a$

(a)	**Step 1**	Add and subtract the numbers	$6 + 2 - 1 = 7$
	Step 2	Answer	$7x$
(b)	**Step 1**	Add and subtract the numbers	$-3 + 7 - 2 = 2$
	Step 2	Answer	$2p$
(c)	**Step 1**	Collect the x terms	$5x - 2x = 3x$
	Step 2	Collect the y terms	$4y - y = 3y$
	Step 3	Answer	$3x + 3y$
(d)	**Step 1**	Collect the a^2 terms	$a^2 + 2a^2 = 3a^2$
	Step 2	Collect the a terms	$4a - a = 3a$
	Step 3	Answer	$3a^2 + 3a$

Practice: simplifying expressions

1.1 Simplify, by collecting like terms.

(a) $3x + 7x - x$

(b) $-4p + 9p - 2p$

(c) $6x + 2y - x - 5y$

(d) $r^2 + 6r + 4r^2 - 2r$

1.2 Simplify, by collecting like terms.

(a) $5y + 2y - 3y$

(b) $-2q + 8q - 3q$

(c) $7x + 5y - 3x - 2y$

(d) $m^2 + 3m + m^2 - m$

1.3 Simplify, by collecting like terms.

(a) $8a + 4a - 5a$

(b) $-5b + 11b - 4b$

(c) $9x + 3y - 4x - y$

(d) $n^2 + 7n + 5n^2 - 4n$

1.4 Simplify, by collecting like terms.

(a) $2m + 9m - 6m$

(b) $-3k + 10k - 6k$

(c) $5x + 6y - 2x - 4y$

(d) $t^2 + 2t + 2t^2 - t$

1.5 Simplify, by collecting like terms.

(a) $10c + 3c - 7c$

(b) $-6d + 12d - d$

(c) $8x + 9y - 6x - 3y$

(d) $x^2 + 8x + 6x^2 - 3x$

1.6 Simplify, by collecting like terms.

(a) $6z + 6z - 4z$

(b) $-w + 7w - 2w$

(c) $4x + 7y - 3x - 6y$

(d) $y^2 + 5y + y^2 - 2y$

Unit 3 · Writing expressions

Worked example: writing expressions from words

(a) Sara is h cm tall. Omar is 6 cm shorter than Sara. Write an expression for Omar's height.

(b) A bag of rice has a mass of 5 kg. Write an expression for the mass of n bags in kg and in grams.

(c) Books cost x dollars each. Pens cost y dollars each. Write an expression for T, the cost of 4 books and 3 pens.

(d) AB = BC = x cm. CD is 2 times as long as AB. DE is 5 cm. Write an expression for the length AE.

(a)	Step 1	Shorter: subtract	$h - 6$
	Step 2	Answer	$h - 6$ cm
(b)	Step 1	n bags: times by n	$5n$ kg
	Step 2	1 kg = 1000 g	$5n \times 1000 = 5000n$ g
(c)	Step 1	4 books, 3 pens	$4x + 3y$
	Step 2	Answer	$T = 4x + 3y$
(d)	Step 1	Add all the parts	$x + x + 2x + 5$
	Step 2	Answer	$4x + 5$ cm

Practice: writing expressions from words

2.1

(a) Bahir is h cm tall. Sienna is 4 cm taller than Bahir. Write an expression for Sienna's height.

(b) A bag of sugar has a mass of 2 kg. Write an expression for the mass of n bags in kg and in grams.

(c) Pens cost x pence each. Rulers cost y pence each. Write an expression for T, the cost of 3 pens and 2 rulers.

(d) AB = BC = x cm. CD is 2 times as long as AB. DE is 3 cm. Write an expression for the length AE.

2.2

(a) Ali is h cm tall. Mona is 9 cm shorter than Ali. Write an expression for Mona's height.

(b) A bag of flour has a mass of 3 kg. Write an expression for the mass of n bags in kg and in grams.

(c) Apples cost x pence each. Oranges cost y pence each. Write an expression for T, the cost of 5 apples and 4 oranges.

(d) AB = BC = x cm. CD is 3 times as long as AB. DE is 7 cm. Write an expression for the length AE.

2.3

(a) Rana is h cm tall. Yusuf is 12 cm taller than Rana. Write an expression for Yusuf's height.

(b) A bag of cement has a mass of 4 kg. Write an expression for the mass of n bags in kg and in grams.

(c) Pencils cost x pence each. Rubbers cost y pence each. Write an expression for T, the cost of 2 pencils and 6 rubbers.

(d) AB = BC = x cm. CD is 4 times as long as AB. DE is 2 cm. Write an expression for the length AE.

Practice: writing expressions from words (continued)

2.4

(a) Hadi is h cm tall. Lina is 7 cm shorter than Hadi. Write an expression for Lina's height.

(b) A bag of potatoes has a mass of 10 kg. Write an expression for the mass of n bags in kg and in grams.

(c) Cakes cost x pence each. Drinks cost y pence each. Write an expression for T, the cost of 6 cakes and 2 drinks.

(d) AB = BC = x cm. CD is 2 times as long as AB. DE is 10 cm. Write an expression for the length AE.

2.5

(a) Nour is h cm tall. Zaid is 15 cm taller than Nour. Write an expression for Zaid's height.

(b) A bag of sand has a mass of 6 kg. Write an expression for the mass of n bags in kg and in grams.

(c) Stamps cost x pence each. Cards cost y pence each. Write an expression for T, the cost of 4 stamps and 5 cards.

(d) AB = BC = x cm. CD is 5 times as long as AB. DE is 1 cm. Write an expression for the length AE.

2.6

(a) Amira is h cm tall. Tariq is 3 cm shorter than Amira. Write an expression for Tariq's height.

(b) A bag of rice has a mass of 25 kg. Write an expression for the mass of n bags in kg and in grams.

(c) Cups cost x pence each. Plates cost y pence each. Write an expression for T, the cost of 7 cups and 3 plates.

(d) AB = BC = x cm. CD is 3 times as long as AB. DE is 8 cm. Write an expression for the length AE.

Unit 3 · Substitution

Worked example: substituting into formulae

$y = mx + c$. Find y when

(a) $m = 5$, $x = 2$, $c = 3$

(b) $m = -3$, $x = -4$, $c = -2$

(c) $T = RH$. Find T when $R = 12$, $H = 6$

(d) $A = x^2 - 7$. Find A when $x = 5$

(a)	**Step 1**	Put the numbers in	$5 \times 2 + 3 = 10 + 3 = 13$
(b)	**Step 1**	Put the numbers in	$-3 \times (-4) + (-2) = 12 - 2 = 10$
(c)	**Step 1**	RH means $R \times H$	$12 \times 6 = 72$
(d)	**Step 1**	Square first	$5^2 - 7 = 25 - 7 = 18$

Practice: substituting into formulae

3.1 $y = mx + c$. Find y when

(a) $m = 8$, $x = 3$, $c = 4$

(b) $m = -2$, $x = -5$, $c = -5$

(c) $T = RH$. Find T when $R = 10$, $H = 40$

(d) $A = x^2 - 5$. Find A when $x = 4$

3.2 $y = mx + c$. Find y when

(a) $m = 6$, $x = 4$, $c = 7$

(b) $m = -4$, $x = -3$, $c = -6$

(c) $T = RH$. Find T when $R = 15$, $H = 8$

(d) $A = x^2 - 9$. Find A when $x = 6$

3.3 $y = mx + c$. Find y when

(a) $m = 3$, $x = 9$, $c = 2$

(b) $m = -5$, $x = -2$, $c = -3$

(c) $T = RH$. Find T when $R = 11$, $H = 7$

(d) $A = x^2 - 10$. Find A when $x = 7$

3.4 $y = mx + c$. Find y when

(a) $m = 7$, $x = 5$, $c = 6$

(b) $m = -3$, $x = -6$, $c = -8$

(c) $T = RH$. Find T when $R = 9$, $H = 30$

(d) $A = x^2 - 4$. Find A when $x = 3$

3.5 $y = mx + c$. Find y when

(a) $m = 4$, $x = 8$, $c = 9$

(b) $m = -6$, $x = -1$, $c = -4$

(c) $T = RH$. Find T when $R = 20$, $H = 12$

(d) $A = x^2 - 12$. Find A when $x = 8$

3.6 $y = mx + c$. Find y when

(a) $m = 9$, $x = 2$, $c = 5$

(b) $m = -2$, $x = -7$, $c = -9$

(c) $T = RH$. Find T when $R = 14$, $H = 5$

(d) $A = x^2 - 1$. Find A when $x = 10$

Unit 3 · Substitution

Worked example: substituting into fractions

$p = 3,\ q = 5,\ r = 3$. Work out

(a) $\dfrac{4p}{r}$ (b) $\dfrac{q^2 - 1}{p}$ (c) $\dfrac{4p + r}{q}$

(a)	Step 1	Work out the top	$4 \times 3 = 12$
	Step 2	Divide top by bottom	$12 \div 3 = 4$
(b)	Step 1	Work out the top	$5^2 - 1 = 24$
	Step 2	Divide top by bottom	$24 \div 3 = 8$
(c)	Step 1	Work out the top	$4 \times 3 + 3 = 15$
	Step 2	Divide top by bottom	$15 \div 5 = 3$

Practice: substituting into fractions

4.1 $p = 3,\ q = 7,\ r = 2$. Work out

(a) $\dfrac{2p}{r}$ (b) $\dfrac{q^2 - 1}{p}$ (c) $\dfrac{4p + r}{q}$

4.2 $p = 5,\ q = 6,\ r = 4$. Work out

(a) $\dfrac{4p}{r}$ (b) $\dfrac{q^2 - 1}{p}$ (c) $\dfrac{4p + r}{q}$

4.3 $p = 7,\ q = 5,\ r = 2$. Work out

(a) $\dfrac{2p}{r}$ (b) $\dfrac{q^2 - 4}{p}$ (c) $\dfrac{4p + r}{q}$

4.4 $p = 4,\ q = 9,\ r = 2$. Work out

(a) $\dfrac{2p}{r}$ (b) $\dfrac{q^2 - 1}{p}$ (c) $\dfrac{4p + r}{q}$

4.5 $p = 8,\ q = 5,\ r = 3$. Work out

(a) $\dfrac{3p}{r}$ (b) $\dfrac{q^2 - 1}{p}$ (c) $\dfrac{4p + r}{q}$

4.6 $p = 9,\ q = 7,\ r = 6$. Work out

(a) $\dfrac{2p}{r}$ (b) $\dfrac{q^2 - 4}{p}$ (c) $\dfrac{4p + r}{q}$

Unit 3 · Expanding brackets

Worked example: expanding a single bracket

Expand

(a) $5(x+2)$ (b) $x(x+6)$ (c) $2k(3k+5)$ (d) Find the value of $2k(3k+5)$ when $k=3$

(a)	**Step 1**	Expand	$5\times x+5\times 2$
	Step 2	Answer	$5x+10$
(b)	**Step 1**	Expand	$x\times x+x\times 6$
	Step 2	Answer	x^2+6x
(c)	**Step 1**	Expand	$2k\times 3k+2k\times 5$
	Step 2	Answer	$6k^2+10k$
(d)	**Step 1**	Put in $k=3$	$2\times 3\times(3\times 3+5)$
	Step 2	Bracket first, then ×	6×14
	Step 3	Answer	84

Practice: expanding a single bracket

5.1 Expand

(a) $4(x+3)$ (b) $x(x+4)$ (c) $3k(2k+4)$ (d) Find the value of $3k(2k+4)$ when $k=2$

5.2 Expand

(a) $6(x+2)$ (b) $x(x+3)$ (c) $4k(3k+1)$ (d) Find the value of $4k(3k+1)$ when $k=2$

5.3 Expand

(a) $3(x+7)$ (b) $x(x+8)$ (c) $5k(2k+3)$ (d) Find the value of $5k(2k+3)$ when $k=1$

5.4 Expand

(a) $7(x+4)$ (b) $x(x+5)$ (c) $2k(4k+7)$ (d) Find the value of $2k(4k+7)$ when $k=3$

5.5 Expand

(a) $8(x+5)$ (b) $x(x+6)$ (c) $6k(2k+1)$ (d) Find the value of $6k(2k+1)$ when $k=2$

5.6 Expand

(a) $9(x+1)$ (b) $x(x+2)$ (c) $3k(5k+2)$ (d) Find the value of $3k(5k+2)$ when $k=2$

Worked example: an expression for the area of a rectangle

A rectangle has a width of y cm. The length is 4 cm longer than the width.

(a) Write an expression for the length. (b) Show that the area is $A = y^2 + 4y$ (c) Work out the area when $y = 6$

(a)	Step 1	Longer: add 4	$y + 4$
	Step 2	Answer	$y + 4$ cm
(b)	Step 1	Area = length × width	$A = y(y + 4)$
	Step 2	Expand	$A = y^2 + 4y$
(c)	Step 1	Put in $y = 6$	$6^2 + 4 \times 6$
	Step 2	Power first, then add	$36 + 24$
	Step 3	Answer	60 cm^2

Practice: an expression for the area of a rectangle

6.1 A rectangle has a width of y cm. The length is 7 cm longer than the width.

(a) Write an expression for the length. (b) Show that the area is $A = y^2 + 7y$ (c) Work out the area when $y = 9$

6.2 A rectangle has a width of y cm. The length is 3 cm longer than the width.

(a) Write an expression for the length. (b) Show that the area is $A = y^2 + 3y$ (c) Work out the area when $y = 8$

6.3 A rectangle has a width of y cm. The length is 5 cm longer than the width.

(a) Write an expression for the length. (b) Show that the area is $A = y^2 + 5y$ (c) Work out the area when $y = 4$

6.4 A rectangle has a width of y cm. The length is 2 cm shorter than the width.

(a) Write an expression for the length. (b) Show that the area is $A = y^2 - 2y$ (c) Work out the area when $y = 10$

6.5 A rectangle has a width of y cm. The length is 9 cm longer than the width.

(a) Write an expression for the length. (b) Show that the area is $A = y^2 + 9y$ (c) Work out the area when $y = 3$

6.6 A rectangle has a width of y cm. The length is 6 cm shorter than the width.

(a) Write an expression for the length. (b) Show that the area is $A = y^2 - 6y$ (c) Work out the area when $y = 11$

Unit 3 · Perimeter expressions

Worked example: writing and using a perimeter expression

A rectangle has a width of x cm. The length is 3 cm longer than the width.
(a) Write an expression for the length. (b) Show that the perimeter is $P = 4x + 6$ (c) Work out P when $x = 5$

(a)	Step 1	Add 3 to the width	$x + 3$
(b)	Step 1	Add all four sides	$x + (x + 3) + x + (x + 3)$
	Step 2	Collect like terms	$P = 4x + 6$
(c)	Step 1	Substitute $x = 5$	$4 \times 5 + 6$
	Step 2	Answer	26 cm

Practice: perimeter expressions

7.1 A rectangle has a width of x cm. The length is 2 cm longer than the width.
(a) Write an expression, in terms of x, for the length of the rectangle.
(b) Show that the perimeter, P, of the rectangle is given by $P = 4x + 4$
(c) Work out the perimeter when $x = 7$

7.2 A rectangle has a width of x cm. The length is 5 cm longer than the width.
(a) Write an expression, in terms of x, for the length of the rectangle.
(b) Show that the perimeter, P, of the rectangle is given by $P = 4x + 10$
(c) Work out the perimeter when $x = 3$

7.3 A rectangle has a width of x cm. The length is 1 cm longer than the width.
(a) Write an expression, in terms of x, for the length of the rectangle.
(b) Show that the perimeter, P, of the rectangle is given by $P = 4x + 2$
(c) Work out the perimeter when $x = 9$

7.4 A rectangle has a width of x cm. The length is 6 cm longer than the width.
(a) Write an expression, in terms of x, for the length of the rectangle.
(b) Show that the perimeter, P, of the rectangle is given by $P = 4x + 12$
(c) Work out the perimeter when $x = 4$

Practice: perimeter expressions (continued)

7.5 A rectangle has a width of x cm. The length is twice the width.

(a) Write an expression, in terms of x, for the length of the rectangle.

(b) Show that the perimeter, P, of the rectangle is given by $P = 6x$

(c) Work out the perimeter when $x = 8$

7.6 Each side of an equilateral triangle is $(x + 3)$ cm long.

(a) Show that the perimeter, P, of the triangle is given by $P = 3x + 9$

(b) Work out the perimeter when $x = 4$